Jalaram Hospital · AfyaSasa Clinical EMR · Confidential operational audit

AfyaSasa EMR

Departmental functionality, workflows, gaps and readiness

Single combined report for hospital management and engineering.
Audit date: 15 October 2026 (Africa/Nairobi)
Source inspected: git `507cc02` on branch `main`
Nature: **read-only inspection and documentation**. No application, database, configuration, or production change.

CODE IMPLEMENTATION
70%
Weighted score of 102 features in source (see method).

VERIFIED THIS AUDIT
0%
No logged-in clinical workflow was exercised.

PRODUCTION VERSION
Unknown
Contabo / live hospital image SHA was not inspected.

How to read the percentages. 70% means: most departmental screens and APIs exist in the current source code, with automated tests from 14 October 2026, but this audit did *not* sit with staff and complete live visits. It is **not** “the hospital is 70% live” and it is **not** “production is 70% on this commit”. Staff have separately reported that OPD is efficient and prescriptions generally work, with some dispensing issues still to investigate.

No patient names, identifiers, credentials, or secrets appear in this document.

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1. How this report was built, and what the percentages mean

1.1 Purpose

This is one document that answers, for every hospital department in AfyaSasa: **what the module is**, **who uses it**, **the actual staff flow as coded**, **what is working in the software**, **what is incomplete or unverified**, and **a readiness percentage with a defined method**.

It consolidates the departmental audit of 15 October 2026. It does not invent a workflow because a menu item exists. A button is not proof that the downstream stock, bill, or result was saved.

1.2 Evidence rules

Claim	Allowed evidence
Exists in software	Exact screen, file, API path, database table in source `507cc02`
Has automated tests	Named Jest spec + dated log of 14 Oct 2026 (380 passed). Tests were not re-run on 15 Oct.
Running locally	Docker containers healthy; `GET /api/v1/health` returned ok. Images were built **before** commit 507cc02.
Used in the hospital	Staff report only, labelled as such. No patient file was opened.
Works in production	**Not claimed.** Production host and version were not inspected.

1.3 Scoring method (denominator = 102 features)

A **feature** is one staff capability with a named screen and/or API (matrix F01–F92 and S01–S10). That is the denominator. Percentages are not guessed.

Status	Meaning	Points	Count
Verified operational	This audit exercised the live workflow	100	0
Implemented, partially verified	Code + tests and/or staff report; no live UAT this audit	75	88
Partially implemented	Essential step missing (device, live HIE, unapplied SQL, mapping-only)	40	10
Configured but unverified	Settings exist; live delivery not proven	45	1
Unverified	Screen exists; this audit did not inspect behaviour	35	1
Blocked or defective	Policy or reproducible block prevents the step	15	1
Not implemented	Confirmed absent	0	1

Hospital code-implementation score = (88×75 + 10×40 + 1×45 + 1×35 + 1×15 + 1×0) / (102×100) = **7,095 / 10,200 = 69.6%, reported as 70%**.

Independently verified operational score this audit = 0 / 102 = **0%**. That is honest: nobody completed a visit, dispense, or payment under auditor observation on 15 October.

Module scores use the same points, only on that module's features. They are comparable to each other. They are **not** "this ward is 75% clinically safe".

1.4 Three percentages you must not mix

1. **70% code implementation** — source at 507cc02 has most departmental functions built.
2. **0% auditor-verified live** — this audit did not log in as staff.
3. **Unknown production alignment** — local Docker images pre-date 507cc02; live Contabo version unknown.

2. Hospital-wide readiness dashboard

Module	Features	Code score	Staff-reported live use	Main gap
Front office	10	72%	Registration / check-in implied by OPD report	Biometrics need a device
OPD	12	75%	**Yes — reported efficient**	Not auditor-verified; consult fee not auto-charged
Pharmacy	14	72%	Prescriptions generally work; **dispensing issues reported**	PH-R1 unreproduced; name-match SKU
Laboratory	10	69%	Not stated this audit	Amend blocked; template sign-off
Radiology	6	75%	Not stated	No PACS
IPD / nursing / ICU / HDU	10	75%	Not stated	Clinical UAT not done
Emergency / theatre / maternity / blood bank	9	67%	Not stated	Blood bank absent; specialty UAT
Finance	9	67%	Not stated	Live M-Pesa / QB unverified; finance metrics uncertified
Store / procurement	6	63%	Not stated	Session/lot SQL unapplied
Documents / reports / marketing	6	58%	Not stated	Email, SMS, KHIS, Director pharmacy KPI
Platform / security / SHA / FHIR	10	65%	Login used in hospital (implied)	Live HIE not verified
Hospital (all 102)	**102**	**70%**	OPD + Rx yes; dispensing mixed	See register



Module arithmetic is in the appendix of this chapter’s scoring notes. OPD 75% is “all twelve features are implemented, partially verified”, not “OPD is three-quarters built”. Staff report OPD is the strongest live pathway.

3. What the system is

3.1 Identity

Item	Value
Product	AfyaSasa Clinical EMR for Jalaram Hospital (tenant code `jalaram`)
Repository	local path Afya-Sasa, branch main
Commit	`507cc02` — message “drugs changes....” — 14 Oct 2026 17:25 EAT
Stack	React frontend, NestJS backend, PostgreSQL (TypeORM schema still named `demo`), Redis, MinIO
API prefix	`/api/v1`
Auth	JWT access token + refresh cookie. Global login guard. Permissions on each API (OR of listed keys). Menu hiding is not security.
Local stack this audit	Containers healthy. Health JSON ok. Images created 14 Oct 2026 11:39 EAT — **older than the commit**.

There is **one inventory ledger** for store and pharmacy. There is **one cashier ledger** (charges and payment transactions). QuickBooks is the accounting general ledger, not a second EMR till.

3.2 Screens staff see (navigation)

36 labelled screens in 13 groups from `frontend/src/lib/navigation.ts`. A user only sees groups they have permission for. The API still enforces permission even if someone crafts a request.

Dashboard: All Queues.
Front office: Register Patient, Patient Registry, Identify Patient, OPD Check-In, Appointments.
OPD: Triage Queue, Doctor Queue, Sick Sheets, Referrals, Orders.
IPD: Inpatient, ICU, HDU, Nursing.
Diagnostics: Laboratory, Lab Queue, Radiology, Imaging Queue.
Specialty: Emergency, Theatre, Maternity.
Pharmacy: Pharmacy, Queue, Dispensing, Sales, Stock, Stock Take, Products, Reports.
Store: Inventory & Store.
Finance: Cashier / Billing, Insurance & Claims.
Documents: Medical Documents, Hospital Library, Reports.
Marketing. Director Dashboard. Control Center and Interoperability.

Not in the menu: blood bank (does not exist), HR/payroll, a separate e-invoice document type.

4. Hospital journey in one picture

> 1. **Front office** searches. If no file, registers the person (duplicate check on identifiers). Then OPD check-in creates today's visit, or a diagnostic-only visit that goes to lab/radiology instead of a doctor.
> 2. **Nurse** triages (vitals). **Doctor** sees own queue, writes SOAP, diagnosis, lab/rad orders, prescription. Completes the encounter.
> 3. **Laboratory / radiology** collect or image, enter results, a verifier publishes. Clinician inbox shows verified work.
> 4. **Pharmacy** sees the prescription as requested. Stock does *not* move until the pharmacist confirms. Then FEFO lots decrease and a pharmacy *charge* is written in the same database transaction. Cash is a later cashier step.
> 5. If the patient is admitted, **IPD** occupies a bed. Nursing records vitals and MAR (MAR is not a stock deduction). Discharge needs a summary. Theatre, maternity, ICU, HDU, emergency are separate workspaces that still depend on the same lab/pharmacy/cashier where orders exist.
> 6. **Cashier** sees charges (owed), takes payment (cash/card/insurance/M-Pesa as configured), reprints a receipt. QuickBooks is supposed to receive accounting later — not verified live.
> 7. **Director** sees counts from the same tables. Finance tiles are not a certified P&L. Pharmacy activity on the director screen is explicitly "data unavailable".

If any of those hops is skipped in real life (for example cashier posts a consult fee that check-in never created, or pharmacy cannot match a drug name), the software still has a path — that is where operational pain appears.

5. Module by module

5.1 Front office and patient registration — code score 72%

IMPLEMENTED IN SOURCE | NOT LIVE-UAT THIS AUDIT

What this module is. The hospital's front door: find a person, create a file if needed, check them into today's service, book appointments, optionally identify by fingerprint if a DigitalPersona device is attached.

Who. Records officers / reception. Permission keys include `patients:create`, `patients:search`, `patients:read`, `encounters:create`, `appointments:read`.

Screens. Register Patient, Patient Registry, Identify Patient, OPD Check-In, Appointments. Header global search.

What it is not. It does not by itself collect money. Check-in shows the clinic consultation fee on screen; it does **not** insert a row into `charges`. It is not SHA eligibility (that is a separate SHA call). Name-only similarity is not treated as a duplicate; unique identifiers are.

> **Actual flow (as coded)**
>
> 1. Staff search `GET /api/v1/patients`. If found, open the file. Do not create a second person.
> 2. If new: `POST /patients/duplicates` then `POST /patients`. Identifiers, next of kin, allergies, chronic conditions can be added on the patient.
> 3. OPD Check-In: choose patient, clinic, visit type, optional doctor, complaint. `POST /opd/encounters`.
> 4. If the patient already has an open OPD visit today (Nairobi day), the API refuses a second one — continue the existing visit.
> 5. Visit type routes the queue token: `new` / `follow_up` / `referral` → doctor (OPD queue). `laboratory_only` or `diagnostic` → laboratory queue. `radiology_only` → radiology queue.
> 6. Appointments: slots, bookings, status, mark arrived (`/appointments`). Hospital-wide "today" list is not automatically doctor-scoped (the doctor screen has a separate "my appointments today").

What is working in the software. Search, register, identifier uniqueness, check-in with visit types, open-visit guard, appointment APIs, patient history/timeline/chart/QR card endpoints. Automated tests exist on patient duplicates and OPD encounter create (prior 14 Oct).

Gaps. Identify Patient depends on biometric hardware and agent — scored partial (40 points). No auditor login. Production unknown.

Readiness. 9 features × 75 + biometrics 40 = 715 / 1,000 = **72%**.

5.2 Outpatient department (OPD) — code score 75%

STRONGEST STAFF-REPORTED MODULE | CODE COMPLETE FOR CORE PATH

What this module is. The clinic visit: nurse triage, doctor consultation, diagnosis, orders, completion, sick sheets, referrals, follow-up.

Who. Nurses (`triage:create`), doctors (`consultations:create` , `diagnoses:create` , `pharmacy:prescribe`), clerks for sick sheets/referrals.

Screens. Triage Queue, Doctor Queue, Sick Sheets, Referrals, Orders. Doctors also prescribe from the consultation workspace, not from the pharmacy menu.

What it is not. It is not the emergency module (ED uses `emergency_encounters`). It is not automatic billing of the consult fee. Completing a visit does not dispense drugs.

> **Actual flow (as coded)**
>
> 1. Encounter exists in status registered (from check-in). Queue type OPD unless diagnostic routing applied.
> 2. Nurse opens Triage Queue, records vitals/category via `POST /opd/encounters/:id/triage` .
> 3. Doctor opens Doctor Queue: `GET /opd/doctor/queue` — own assigned patients plus unassigned. Supervisors see all. Plus `GET /opd/doctor/appointments/today` .
> 4. Consultation: SOAP `POST .../consultations` . Diagnoses (ICD text in EMR; no live ICD-10 web service). Notes and file attachments.
> 5. Orders: laboratory request and/or radiology request and/or multi-line prescription `POST /clinical-orders/pharmacy/prescription` . Prescription status becomes *requested*. Stock is untouched.
> 6. Optional follow-up date. Complete visit `PATCH .../status` through the encounter workflow service (illegal status jumps are rejected).
> 7. Sick sheet and referral workspaces are separate documents/letters, not a substitute for the consultation note.

What is working. Staff report the pathway is efficient. Source implements the full chain above. Queue scoping is unit-tested. Prescription grouping is unit-tested.

Gaps. This audit did not watch a doctor complete a visit. Diagnostic-only patients can still receive a consult charge later if cashier adds one. Follow-up SMS is not evidenced as live.

Readiness. 12/12 implemented, partially verified = **75%**. Combined with staff report, this is the module management can treat as the operational backbone — still not 100% auditor-verified.

5.3 Pharmacy and drug dispensing — code score 72%

PRIORITY MODULE | PRESCRIBE REPORTED OK | DISPENSE ISSUES REPORTED, NOT REPRODUCED

What this module is. The place medicines leave the shelf. It uses the **same** stock tables as the store. Location code `PHARMACY` . Confirming a prescription writes a `DISPENSE` ledger line and, if priced, a `charges` row in the **same database transaction**. Payment is still cashier.

Who. Pharmacists (`pharmacy:read` , `pharmacy:dispense`). Doctors prescribe; they should not confirm stock.

Screens. Pharmacy home, Queue, Dispensing, Sales, Stock, Stock Take, Products, Reports. OTC panel for walk-in sales.

What it is not. Nursing MAR is not dispensing. Director “pharmacy activity” is not this module (that KPI is unavailable). Store requisition is how bulk stock arrives at PHARMACY, not how a patient is served.

> **Actual flow (as coded)**
>
> 1. Doctor already created a prescription header + lines (`clinical_orders` , status requested, qty dispensed 0).
> 2. Pharmacist loads queue `GET /clinical-orders` (pharmacy module). UI groups lines by prescription group.
> 3. UI matches each line to a catalogue item: prefers `metadata.itemId` ; if missing, **normalised name equality**. That heuristic can pick the wrong SKU if two products share a name and itemId was not stored.
> 4. Pharmacist reviews quantities and ticks confirm. API `POST /inventory/dispense/prescription` with `confirm: true` . Without confirm, API rejects.
> 5. Backend locks the order row, allocates non-expired FEFO batches, reduces `qtyOnHand` , writes `inventory_transactions` type DISPENSE, upserts pharmacy charge using a `dispenseOpId` so the same operation should not double-bill if retried correctly.
> 6. If stock is short, expired, cancelled, or already fully dispensed — error, no silent partial without remaining-qty rules. Partial dispense is supported via remaining quantity metadata.
> 7. If charge insert fails, the whole transaction rolls back — staff see an error; stock should remain. Billing retry exists without moving stock again: `POST /inventory/dispense/retry-billing/:orderId` .
> 8. Patient is told to pay at cashier. History filters show dispensed / partial.

OTC. `POST /inventory/dispense/otc` . Frontend sends a random `saleId` per submit (idempotent for that click). Walk-in without patient is unbilled by design. Without saleId the API would mint a new id each call (not idempotent).

What is working in software. Confirm gate, FEFO, same-txn charge, order lock, insufficient-stock errors, cancelled-Rx refusal, stock-take workspace, catalogue, low/expiry views. Strong unit tests and 12 disposable-Postgres concurrency tests on 14 Oct (miniature schema, not hospital data).

What staff reported. Prescriptions generally work. A few dispensing problems remain. This audit **did not dispense** and therefore **confirmed zero dispensing bugs**. The report is an open P1 until someone captures the exact error (wrong drug, no stock movement, double charge, timeout, permission).

Likely technical suspects (not proven incidents): name-match SKU; running Docker images older than the stock-integrity commit; concurrent stock-take without the planned unique session table.

Readiness. 13×75 + sales-view 35 = 1,010 / 1,400 = **72%**. Do not read this as “dispensing is 72% reliable in the cubicle”.

5.4 Laboratory — code score 69%

What it is. Test catalogue, requests from OPD/IPD, sample + barcode, receive/reject, result entry with flags, verification by a different permission, reports, clinician inbox, critical results.

Who. Lab technicians enter; lab verifiers verify (`lab_results:verify`). Cancel needs `lab_requests:cancel` — not on the technician template. Doctors create requests.

> 1. Request `POST /laboratory/requests` (status requested).

2. Collect sample, barcode generate/scan, receive or reject.
3. Enter results (panel or item). Missing reference ranges → flag “unassessed”, not a fake normal.
4. Verifier `POST /laboratory/requests/:id/verify`.
5. Report PDF/layouts. Clinician inbox / critical list.
6. Amend of a verified result exists in code but is **switched off** (`AFYASASA_LAB_AMENDMENT_ENABLED=false` on compose). That is policy, scored as blocked (15 points), not a random crash.

Gaps. Lab report templates still need clinical sign-off. Isolated lab Postgres tests (10 passed 14 Oct) used a toy schema. No live sample was processed this audit.

Readiness. 9×75 + amend 15 = 690 / 1,000 = **69%**.

5.5 Radiology and imaging — code score 75%

What it is. Modalities catalogue, requests, status (requested → scheduled → in_progress), report + verify, file attachments, clinician inbox. Diagnostic-only OPD visits land on the radiology queue.

What it is not. A PACS / DICOM viewer. Attachments are files in object storage.

1. Create request `POST /radiology/requests`.
2. Move status; perform study; attach files if any.
3. Write report; verify. Review endpoint exists for downstream clinicians.

Readiness. 6×75 = **75%** implementation. Scheduling depth is limited compared with a full RIS. Live imaging UAT not done.

5.6 Inpatient, nursing, ICU, HDU — code score 75%

What IPD is. Wards and beds, admit (bed occupied), ward transfer, progress notes, discharge summary + PDF, discharge that **requires** a summary, cancel admission.

What nursing is. Vitals, MAR (medication administration record), observations, shift notes. MAR documents that a dose was given. It does **not** reduce pharmacy stock. Inpatient medicines still go through pharmacy confirm if prescribed.

What ICU / HDU are. Separate modules, wired in the current app (the August 2026 gap note that ICU UI was unwired is stale). ICU: admissions, observations, ventilator, fluids, rounds. HDU: admissions, observations, rounds, status.

1. Admit `POST /inpatient/admissions` — bed taken.
2. Care: notes, nursing, orders to lab/rad/pharmacy as for OPD.
3. Optional transfer between wards.
4. Accommodation charging is a *job* using catalogue rates; empty rate means no charge appears.
5. Write discharge summary, then discharge. PDF available.

Readiness. 10×75 = **75%**. Clinical UAT of ICU/HDU/discharge not done this audit.

5.7 Emergency — coded, UAT unverified (in the 67% specialty group)

What it is. ED command center: register, dashboard, queue, bays, alerts, triage, notes, observation, disposition.

What it is not. An OPD encounter. Tables are `emergency_encounters`. Disposition to admit/discharge/refer is in the API; walking a real patient from ED into IPD was not observed.

> 1. `POST /emergency/register` → triage → assign bay → notes/observation → `POST /emergency/:id/disposition`.

5.8 Theatre — coded, UAT unverified

Rooms, procedure catalogue (including import), bookings, status, staff assignment, notes, complications. Consumables are expected via store requisitions, not automatic theatre stock deduction. Tests: theatre service and notes specs (prior 14 Oct).

5.9 Maternity — coded, UAT unverified

Pregnancies, ANC visits, labour records, partograph, deliveries, postnatal, newborns, optional `register-newborn-patient`, mother-baby registry, maternity units. **Risk:** registering a newborn must not duplicate the mother's patient file. Clinical UAT not done.

5.10 Blood bank — code score 0% of that feature

NOT IMPLEMENTED No screen, no API, no table. Do not store blood units as pharmacy lots. If transfusion is a live hospital process, it is outside AfyaSasa today. Combined with ED/theatre/maternity this slice scores **67%** because eight specialty features are implemented and blood bank is zero.

5.11 Billing, payments, finance — code score 67%

What it is. The hospital till: catalogue of prices, charges (what is owed), payments (what was collected), refunds, cashier summary and close, exceptions, IPD accommodation job, statement, M-Pesa STK + callback, QuickBooks queue/Web Connector, insurance claim *drafts*.

What it is not. An invoice module (no separate invoice table found). Not QuickBooks itself. Not "revenue" on the director screen (collections ≠ revenue ≠ GL).

> 1. A **charge** appears because pharmacy dispense created it, cashier added a manual/catalogue charge, or accommodation job ran with a priced rate. OPD check-in does not create the consult charge.
> 2. Cashier looks up patient outstanding `GET /payments/outstanding` and charges.
> 3. Collect: manual payment (cash/card/insurance as posted) or M-Pesa STK. Callback is IP-allowlisted. **Do not test STK on a real phone from this audit.**
> 4. Receipt is a reprint of a completed payment, not a third ledger.

> 5. Refund creates a reversing payment row (metadata kind refund). Original payment remains.
> 6. QuickBooks: queue + web connector. Analytics show accounting as pending. Tests reject missing QBWC credentials.

Readiness. 7×75 + M-Pesa 40 + QuickBooks 40 = 605 / 900 = **67%**. Director finance is not certified for management decisions until Finance completes a period worksheet (billed vs collected vs outstanding vs QB).

5.12 Procurement, inventory, store — code score 63%

What it is. The stock truth: items, batches, locations, ledger. Goods receipt, requisition approve/issue/acknowledge, transfer ship/receive, stock take (posts *variance*, not "counted quantity as a receipt"), bulk price, import preview/confirm, integrity panel, low-stock alerts, barcode.

What it is not. A second pharmacy database. Pharmacy PHARMACY location is one location on this ledger. Buying list lives in the inventory UI; a separate Supplier company module was not found as its own controller.

> 1. Receive goods → RECEIPT on batches.
> 2. Issue from store to pharmacy (requisition) or transfer between locations.
> 3. Usable qty at PHARMACY is what dispensing can allocate (FEFO, not expired).
> 4. Stock take: import count sheet, preview, confirm. Concurrent posts currently rely on advisory locks; a durable unique session table is **planned SQL, not applied**.

Readiness. 4×75 + two unapplied SQL features ×40 = 380 / 600 = **63%**.

5.13 Documents, reports, marketing — code score 58%

Medical documents linked to the patient, hospital library (policies/forms), reports hub (OPD summary, admissions, discharges, occupancy, emergency stats, disease register, department revenue, MOH 705/706/717 mapping and docx, lab/theatre/maternity/referral/ICU report endpoints), marketing outreach log.

Email of reports and password reset: admin email settings exist; 14 Oct tests showed mail not configured / allowlist / degraded. Scored configured-but-unverified (45). **SMS bulk** exists — do not treat as live patient SMS (40). **MOH/KHIS submit** is mapping, not a live ministry post (40). Director pharmacy KPI is missing inside analytics (the analytics feature is still 75 because the rest of the dashboard is built).

Readiness. 3×75 + 40 + 45 + 40 = 350 / 600 = **58%** — lowest departmental slice because delivery and government submit are incomplete.

5.14 Director dashboard — inside reports, finance not certified

Screen: Director Dashboard → `GET /reports/executive-analytics`. Definitions live in `docs/AFYASASA_DASHBOARD_METRIC_DEFINITIONS.md`.

Number on screen	What it actually is	Use for decisions?

OPD visits	Count of OPD encounters in the date range	Directional if check-in is reliable
New patients	Registrations, not visits	Do not confuse with attendance
ED / admissions / discharges / occupancy	From ED table, admissions, beds	Occupancy unavailable if zero beds mastered
Lab/rad pending	Point-in-time worklist	Backlog, not quality
Lab TAT	Average hours among verified only	Blank if nothing verified
Pharmacy activity	**Data unavailable**	Do not invent from stock movements
Billed / collected / refunds / outstanding	Charges vs payments vs point-in-time owed	**Not certified; not GL**
Accounting	Pending QuickBooks	Do not call EMR cash “deposits”

Finance tiles hide unless director/admin/superadmin or payments/settings permission. Doctor fixtures received 403 in tests — by design.

5.15 Users, roles, audit, control center — platform score 65% with SHA/FHIR

Login, refresh, logout, forgot/reset password, MFA endpoints, session policy. Admin: users (create, patch, activate, deactivate, unlock, reset password, assign roles and departments), roles, permissions, templates, clinics, departments, settings, system health, backups list (restore endpoint exists — do not invoke casually), audit logs export/import, PHI access report.

Permission model is **OR**: any listed key grants the route. Menu hide is cosmetic. Audit interceptor records HTTP mutations; it is not a full old-value/new-value clinical snapshot of every column.

SHA: status, eligibility post, coverage read — local mapping, not a live SHA portal. FHIR: map patient/encounter, synthetic validate — not a FHIR server. Insurance: payers and claim drafts; portal claims stay on sha.go.ke per product copy.

Readiness. 7×75 + SHA 40 + FHIR 40 + shared-report delivery 40 = 645 / 1,000 = **65%**.

6. OPD and pharmacy — the special investigation

Management observation given to this audit: OPD efficient; prescriptions generally fine; dispensing has had a few issues. Treated as **user report**, not a passed test programme.

Hop	Record	Expected	Coded behaviour
Register / check-in	patients, encounters	One open OPD visit per day	Yes; diagnostic types skip doctor
Consult	consultations, diagnoses	SOAP + optional ICD	Yes
Prescribe	clinical_orders	requested, stock unchanged	Yes
Queue	same orders	Pharmacist sees lines	Yes
Match drug	UI	Correct SKU	itemId else name — risk
Confirm	batches + ledger + charge	One atomic write	Yes if confirm true and stock exists
Pay	payment_transactions	Later, cashier	Yes, separate

Independently confirmed dispensing defects: none (no live Rx touched).

Open register: PH-R1 reported P1; PH-S1 name match; PH-S2 charge uniqueness at app layer; PH-S3 unapplied SQL; PH-S4 image lag vs commit.

Next diagnostic (when authorised): one synthetic prescription, capture API error body, remaining qty, batch qty, ledger, charge. Do not use a live patient if avoidable. Do not rewrite OPD or FEFO while investigating.

7. Money words (do not mix these)

Word	Table	Meaning
Charge	`demo.charges`	Amount owed. Not cash.
Payment	`demo.payment_transactions`	Collection or refund.
Receipt	Print of a payment	Not a fourth ledger.
Invoice	Not found as its own entity	Do not assume e-invoicing.
General ledger	QuickBooks	EMR is the operational till.

Outstanding on a patient is owed − paid − waived at *query time*. It is not “this month billed minus this month collected”. Mixing those is how management reports go wrong.

8. Integrations — maturity, not slogans

Scale: 1 absent · 2 documented · 3 coded locally · 4 automated tests · 5 sandbox proven · 6 production config seen · 7 live end-to-end proven. This audit never claimed 5–7 for any row. Credentials are not printed here.

Integration	Level	What exists	What does not
SHA / DHA	3	`/sha/status`, eligibility, coverage	Live cover decision unverified
FHIR	3	Map patient/encounter	Not a FHIR server; sandbox disconnected until official credentials
Insurance drafts	3	Payers, claim drafts	Portal submission
M-Pesa	3–4	STK + callback + specs	Live STK
QuickBooks	3–4	Queue, QBWC	Live connected books
Email	3 + unverified	Admin UI, mail service	Proven delivery of PHI reports
SMS	3 partial	Bulk + logs	Live patient SMS programme
Biometrics	3 partial	Identify workspace	Without DigitalPersona agent
MinIO files	3	Local container healthy	—
PACS	1	—	No DICOM archive
Blood bank LIS	1	—	Entire module
KHIS	2–3	MOH form mapping / docx	Ministry submit

9. Issues and next work (no fixes in this audit)

ID	Type	Severity	Plain language
PH-R1	Reported, unreproduced	P1	Pharmacists have had dispensing problems. Cause unknown until captured.
PH-S1	Suspected	P1 suspected	Matching drugs by name can pick the wrong product.
PH-S4	Operations	P1 local	Running local images may not include the 14 Oct stock-integrity commit.
PH-S2 / S3	Suspected	P2	Charge uniqueness and unique lot/session SQL are application-level / unapplied.
LAB amend	Policy block	P2	Verified results cannot be amended while the flag is false — usually correct.
Finance certify	Process	P2	Do not steer the hospital on uncertified billed/collected tiles.
SHA / M-Pesa / QB / mail	Unverified live	P2	Coded; not proven in production by this audit.
Blood bank / PACS / pharmacy KPI	Missing	P3	Not required to keep OPD running; do not fake them.

No P0 was assigned: no proven wrong-drug event, no proven duplicate cash, no proven break-in, no proven stock corruption during this read-only audit. If PH-R1 turns out to be wrong medication issued, that becomes P0 immediately.

Preserve OPD, prescribe, FEFO allocate, payment posting. Next authorised phases: (A) synthetic dispense diagnosis, (B) align local images when IT agrees, (C) read-only duplicate-lot snapshot before any unique index, (D) finance worksheet, (E) lab template sign-off, (F) sandbox integrations only.

10. Tests, evidence, limitations

10.1 Tests (14 Oct 2026 17:20 EAT — not re-run 15 Oct)

Log: `docs/release/2026-10-14-test-evidence.log`. Tree was dirty on parent `a75c2e4`, then committed as `507cc02`.

- Disposable Postgres inventory concurrency: 12 passed.
- Disposable Postgres laboratory: 10 passed.
- Full backend Jest: 71 suites, **380 passed, 0 failed**.
- Frontend and backend TypeScript compile: exit 0.
- Mail during tests: not configured / allowlist / degraded then a mock accept. QBWC: credentials missing rejected. Auth: 401 without token; 403 doctor on director analytics.

Those tests prove code behaviour in isolation. They do not prove the live pharmacy window at Jalaram.

10.2 What this work did not do

- No production access, no migrations, no seeds, no container restarts, no deploys, no commits, no patient data, no real payments, SMS, email, or SHA calls.
- No logged-in click-through of any department on 15 October.
- Did not claim local Docker equals production.

10.3 Main APIs (all under /api/v1)

`auth`, `admin`, `admin/email`, `patients`, `opd`, `appointments`, `inpatient`, `emergency`, `nursing`, `icu`, `hdu`, `laboratory`, `laboratory/catalog`, `radiology`, `theatre`, `maternity`, `clinical-orders`, `inventory`, `payments/*`, `reports`, `exports`, `documents`, `storage`, `referrals`, `queue`, `marketing`, `notifications`, `biometrics`, `sha`, `integrations/fhir`, `insurance`, `health`.

10.4 Bottom line for the Director

AfyaSasa already covers the departments a hospital runs day to day — registration, OPD, diagnostics, wards, theatre, maternity, pharmacy, store, cashier, reports, admin — in source code, with a large automated test suite from 14 October 2026.

Treat OPD as the live clinical spine based on staff report, not based on a new auditor sit-in.

Treat pharmacy dispensing as the open operational risk until one captured failure (or a clean synthetic success) explains the staff complaints.

Treat director finance numbers as operational, not audited accounts. QuickBooks remains the books.

Hospital code-implementation readiness: 70%. Auditor-verified live readiness this audit: 0%. Production version: unknown. Those three sentences are the accurate summary; anything smoother would be marketing, not an audit.

Supporting markdown pack (same findings, split files): docs/audits/afyasasa-departmental-audit/README.md and 01–09. This PDF is the single combined narrative requested for management.